

CSE331: Automata and Computability

Assignment 2

Deadline: 12/08/2026 11:59 PM

Show the following language is not regular using pumping lemma

1. $L = \{ 0^{2n}1^n \mid n \geq 0 \}$, where $\Sigma = \{0, 1\}$.
2. $L = \{ w \mid w \text{ is not a palindrome} \}$, where $\Sigma = \{0, 1\}$.
3. $L = \{ a^m b^n \mid m \neq n \}$, where $\Sigma = \{a, b\}$.
4. $L(M) \rightarrow \{ 0^i 1^j 2^k \mid i, j, k \geq 0 \text{ and if } i = 1 \text{ then } j = k, \text{ otherwise } j \neq k \}$

Context free grammar

1. $L = \{ w \in \{0,1\}^* : w = 0^n 1^m, n, m \geq 0 \text{ and } 2n = 3m \}$
2. $L = \{ w \in \{0,1\}^* : \text{the length of } w \text{ is divisible by } 3 \}$
3. $L = \{ w \in \{0,1\}^* : w_1 \# w_2, \text{ the number of } 0\text{'s in } w_1 \text{ is equal to number of } 1\text{'s in } w_2 \}$
4. $L = \{ w \in \{0,1,2\}^* : w = 0^i 2^j 1^k, \text{ where } i = k, i, k \geq 1 \text{ and } j \geq 2 \}$
5. $L = \{ w \in \{0,1,2\}^* : w = 0^i 2^j 1^k, \text{ where } i \text{ is multiple of two, } k \text{ is two more than multiple of } 3, j = k+i \text{ and } i, j, k \geq 0 \}$
6. $L = \{ w \in \{0,1\}^* : w \text{ is not a palindrome} \}$
7. $L = \{ w \# x \mid w, x \in \{0, 1\}^* ; w^R \text{ is a substring of } x \}$

Derivation, Parse tree and Ambiguity

Using the string 'abaabbbba' show that the following grammar is ambiguous:

$S \rightarrow PbQ$

$P \rightarrow aP \mid bP \mid a \mid b$

$Q \rightarrow aQa \mid aQb \mid bQa \mid bQb \mid \epsilon$

Push down automata

1. $L = \{ w \in \{0,1,2\}^* : w = 0^i 2^j 1^k, \text{ where } i = k, i, k \geq 1 \text{ and } j \geq 2 \}$
2. $L = \{ w \in \{0,1,2\}^* : w = 0^i 2^j 1^k, \text{ where } i \text{ is multiple of two, } k \text{ is two more than multiple of } 3, j = k+i \text{ and } i, j, k \geq 0 \}$
3. $L = \{ w \in \{0,1\}^* : \text{the number of } 0\text{'s and } 1\text{'s are different in } w \}$
4. $L = \{ w \# x : w, x \in \{0,1\}^* \text{ and } w^R \text{ is a substring of } x \}$
5. Recall that for a string w , $|w|$ denotes the length of w . $\Sigma = \{0,1\}$
 $L1 = \{ w \in \Sigma^* : w \text{ contains exactly two } 1\text{'s} \}$
 $L2 = \{ x \# y : x \in \Sigma^*, y \in L1, |x| = |y| \}$
Construct a PDA for $L2$.

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Turing Machine

A.

$L(M) \rightarrow \{w \in \{0, 1\}^* \mid w \text{ contains equal numbers of 0s and 1s}\}$, where $\Sigma = \{0, 1\}$

B.

$L(M) \rightarrow \{0^n 1^n 2^n \mid n \geq 0\}$

C.

$L(M) \rightarrow \{0^i 1^j 2^k \mid i, j, k > 0 \text{ and } k = i * j\}$

D.

$L(M) \rightarrow \{0^{2^n} \mid n \geq 0\}$

E.

$L(M) \rightarrow \{w \in \{0, 1\}^* \mid w \text{ is a palindrome}\}$

Prove that the following languages are decidable:

A.

$L = \{\langle G \rangle \mid G \text{ is a connected undirected graph}\}$

B.

$L = \{\langle A, B \rangle \mid A \text{ and } B \text{ are DFAs and } L(A) = L(B)\}$

Show that the following statements are true

A. The collection of decidable languages is closed under Union.

B. The collection of decidable languages is closed under Complement.

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Reducibility

Given $A_{TM} \rightarrow \{\langle M, w \rangle \mid M \text{ is a TM that accepts } w\}$ is undecidable, prove that the following languages are undecidable.

A.

$RL_{TM} \rightarrow \{\langle M, w \rangle \mid M \text{ is a TM that accepts } w\}$

B.

$M_{000} \rightarrow \{\langle M \rangle \mid M \text{ is a TM that accepts } 000\}$